

EMS Multi-Agent POC Backend Handoff

A concise backend handoff for the EMS chatbot, 8091 multi-agent proof of concept, pgvector RAG storage, and MCP integration point.

Running URLs

- Normal chatbot UI: `http://127.0.0.1:8085/`
- Multi-agent POC webpage: `http://127.0.0.1:8091/`
- Multi-agent POC API: POST `http://127.0.0.1:8091/multi-agent-chat`
- Health check: GET `http://127.0.0.1:8091/health`

Docker Services

```
cd C:\chatbot-stack\Chatbot-UI
docker compose up --build -d
```

- ollama: local model runtime.
- ollama-init: pulls required models.
- postgres: PostgreSQL + pgvector.
- chatbot-api: backend API for 8085.
- agent-poc-api: backend/webpage/API for 8091.
- chatbot-ui: nginx frontend for 8085.

Models To Pull

```
ollama pull qwen2.5:3b
ollama pull deepseek-r1:7b
ollama pull nomic-embed-text
```

- OLLAMA_MODEL=qwen2.5:3b
- DEEPSEEK_MODEL=deepseek-r1:7b
- CHAT_MODEL=deepseek-r1:7b
- EMBEDDING_MODEL=nomic-embed-text
- 8091 Agent 1: qwen2.5:3b
- 8091 Agent 2: qwen2.5:3b
- 8091 final decision maker: deepseek-r1:7b

Multi-Agent API

```
POST http://127.0.0.1:8091/multi-agent-chat
Content-Type: application/json
```

```
{
  "message": "My Janitza UMG 509 shows voltage sag and one feeder is near rated current. What are the likely causes and s"
}
```

Response Shape

```
{
  "reply": "Final compact answer from DeepSeek",
  "provider": "multi-agent-poc",
  "model": "deepseek-r1:7b",
  "decision_model": "deepseek-r1:7b",
  "is_ems_related": true,
  "sources": [
    {
      "title": "Janitza UMG Power Quality Knowledge",
      "standard_name": "Janitza UMG / EMS Power Quality",

```

```
    "score": 0.53
  },
  ],
  "agent_discussion": [
    {
      "agent": "Agent 1 - Ollama EMS Triage",
      "role": "Quickly classify the issue and identify immediate EMS checks.",
      "model": "qwen2.5:3b",
      "answer": "Agent 1 answer"
    },
    {
      "agent": "Agent 2 - Qwen Power Quality",
      "role": "Find likely electrical root causes and UMG meter readings to inspect.",
      "model": "qwen2.5:3b",
      "answer": "Agent 2 answer"
    }
  ],
  "qa_log_id": "uuid"
}
```

Current System Flow

```
8091 webpage
-> POST /multi-agent-chat
-> EMS keyword guard
-> pgvector document retrieval
-> Agent 1 with qwen2.5:3b
-> Agent 2 with qwen2.5:3b
-> DeepSeek final decision maker
-> Save Q&A into qa_logs/chat_messages
-> Save question embedding into qa_embeddings
-> Return final answer, agent discussion, and sources
```

EMS Safety Prompt Rules

- Match the user's language exactly. English in, English out. Malay in, Malay out.
- Reject adding load or overriding alarms when hardware limits are exceeded.
- Never suggest tolerant language when hardware is above 100% rated capacity.
- For critical limit breach, use DIRECT REJECTION, PHYSICAL REASONING, and MITIGATION ACTION.

Database / pgAdmin

```
Host: 127.0.0.1
Port: 15432
Database: chatbot
Username: chatbot
Password: chatbot
```

- documents: uploaded/seeded knowledge documents.
- document_chunks: vector chunks for RAG library search.
- chat_sessions: chat sessions.
- chat_messages: user/assistant message history.
- qa_logs: final Q&A; logs with source metadata.
- qa_embeddings: vector embeddings of EMS-related user questions.
- feedback: future rating/comment table.

Knowledge Ingestion API

```
POST http://127.0.0.1:8091/knowledge
Content-Type: application/json
```

```
{
  "title": "UMG 509 Voltage Sag Notes",
  "standard_name": "Janitza UMG / Power Quality",
  "source_type": "manual",
  "text": "Voltage sag may be caused by motor starting, overloaded feeder, loose terminal, phase imbalance, transformer e
}
```

MCP Integration Point

Call the MCP server after EMS guard and pgvector retrieval, before the role-agent prompts. The MCP data should be passed into Agent 1, Agent 2, and the DeepSeek final decision maker.

```
User question
-> EMS guard
-> Retrieve document context from pgvector
-> Call MCP server for live EMS/device data
-> Agent 1 receives document context + MCP data
-> Agent 2 receives document context + MCP data
-> DeepSeek final decision maker receives summaries + MCP data
-> Save final Q&A and embeddings
```

Recommended MCP Data Contract

```
{
  "device_id": "UMG-509-MSB-01",
```

```

"device_model": "Janitza UMG 509",
"timestamp": "2026-09-09T14:00:00+08:00",
"measurements": {
  "voltage_l1": 230.1,
  "voltage_l2": 184.2,
  "current_l2": 210.0,
  "rated_current": 200.0,
  "frequency": 50.0,
  "thd_v": 4.2,
  "power_factor": 0.86
},
"alarms": [
  {"type": "voltage_sag", "phase": "L2", "severity": "critical"}
]
}

```

Useful SQL

```

SELECT question, answer, is_ems_related, created_at
FROM qa_logs
ORDER BY created_at DESC
LIMIT 10;

```

```

SELECT question, left(answer, 160) AS answer_preview, created_at
FROM qa_embeddings
ORDER BY created_at DESC
LIMIT 10;

```

```

SELECT d.title, d.standard_name, left(dc.chunk_text, 160) AS chunk_preview
FROM document_chunks dc
JOIN documents d ON d.id = dc.document_id
ORDER BY dc.created_at DESC
LIMIT 10;

```